

ADITHI LOKA

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PROFESSIONAL SUMMARY

Software Engineer specializing in distributed data systems, MLOps, and lakehouse architectures at retail scale. Led high-impact initiatives across streaming, Spark, Kafka, Cassandra, BigQuery, feature orchestration, and cloud cost optimization, including \$1M+ annual savings, petabyte-scale transfer reduction, and automated ML retraining systems delivering \$900K+ additional annualized fraud impact.

TECHNICAL SKILLS

Distributed Data Systems: Apache Spark, Apache Flink, Kafka, Hadoop, Hive, Beam/Dataflow, GCP Pub/Sub, Dataproc, Airflow, CI/CD

Lakehouse & Modeling: BigQuery, BigTable, Oracle, Cassandra/NoSQL, GCS, Iceberg/Hudi, dimensional modeling, slowly changing dimensions, warehouse design

Data Quality & Governance: data contracts, point-in-time correctness, lineage, metadata management, schema evolution, lifecycle/retention, privacy-preserving PII handling

ML Engineering: Feast, vector-based model serving, MLflow, model monitoring, XGBoost, VAE models (Variational Autoencoders), TensorFlow, Scikit-Learn, AutoML, LLM/RAG integration

Languages, BI & Cloud: Python, Scala, Java, SQL, GCP, AWS, Docker, Kubernetes, Looker, warehouse performance tuning

PROFESSIONAL EXPERIENCE

SAM'S CLUB (WALMART), USA | Senior Software Engineer - Data and ML

Nov 2024 - Present

- Owned a \$4.5M+ Cloud Cost Champion roadmap; authored architecture decision documents and standards adopted across 6 engineering teams and delivered \$1M+ annual savings.
- Architected a 100TB+ event-driven Medallion Lakehouse on GCS with lifecycle, retention, and archival policies; reduced Cassandra disk usage 44%.
- Designed ML feature orchestration for fraud systems using BigQuery and Feast with data contracts and point-in-time correctness; reduced feature latency 60%.
- Architected Spark-based distributed processing for a vector-based model serving layer alongside Feast; reduced model-serving transfer payloads from 40GB to 4GB and avoided 1PB+ annual data transfer.
- Spearheaded MLOps automation for DS-owned Fuel and SNG fraud models across data science, platform, and business stakeholders; enabled on-demand model refreshes, 46% more chargeback capture, and \$900K+ incremental annualized savings.
- Managed Kubernetes-hosted Flask model-serving APIs for 700K+ daily fraud transactions; deployed model PKL artifacts, integrated Cassandra ingestion, and supported A/B testing, p95/p99 monitoring, load testing, and autoscaling.

SAM'S CLUB (WALMART), USA | Software Engineer

Apr 2023 - Oct 2024

- Designed caching architecture that reduced cloud ingress traffic 82% from 45 GBPS to under 9 GBPS; tuned Spark/Dataproc and warehouse workloads to save \$100K annually.
- Modernized Databricks-backed Power BI reporting workloads into GCP Dataproc clusters, Jupyter notebooks, and Looker dashboards; standardized catalog definitions for high-impact dashboards sourcing hot-storage data.
- Created dimensional fraud analytics models across core business entities, model scores, and chargeback outcomes; introduced a masked pre-production layer with strict access controls so daily fraud experiments never handled raw PII.
- Re-architected retail-channel batch processes into streaming pipelines, reducing p95 latency by approximately 100ms and enabling minute-level freshness.

WALMART INC, USA | Software Engineer

Jul 2021 - Mar 2023

- Led Hadoop-to-GCP/Airflow migration with CI/CD deployment controls and monitoring for 100+ production workflows with 99.5% uptime, reducing MTTD from 3 days to under 3 hours.
- Built Scala, Kafka, and BigQuery ingestion pipelines with inventory data lineage and rule-based anomaly detection, doubling downstream data accuracy and improving reporting reliability.

ACCENTURE SOLUTIONS, INDIA | Software Engineer

May 2018 - Jul 2019

- Built Spring Boot REST APIs and ETL workflows using Azure Data Factory and SSIS to improve data exchange across enterprise systems.

EDUCATION & CERTIFICATIONS

MS in Computer Science, University of Colorado Denver, GPA 4.0 | Google Cloud Certified Professional Data Engineer

AWARDS & ACHIEVEMENTS

Two-Time Exemplary Performer (2023, 2025); MAD Award for successful migration; Bravo Award for fraud detection/cost optimization; Outstanding Graduate Student Award, UCD Computer Science (Spring 2021).